

Privacy, Economics, and Price Discrimination on the Internet

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Motivation: How to explain two puzzles:

- Persistent erosion of privacy in spite of strong public concern
- Disappointing record of shopping bots, auctions, micropayments, DRM, ...

Main points:

- Common explanation: price discrimination
- Incentives to price discriminate are increasing
- Ability to price discriminate is improving
- Price discrimination often arises in competitive markets, as a byproduct of scale effects, and not of “market power” and monopoly
- Behavioral economics will constrain price discrimination and drive it into hidden forms
- Government role in ecommerce likely to be substantial but ambiguous

Economic desirability of price discrimination:

Zach: willing to prepare a report on digital cash for \$1,500

Alice: willing to pay \$700

Bob: willing to pay \$1,000

Uniform pricing makes transaction impossible

Charging Alice \$650 and Bob \$950 makes everybody better off (in conventional economic model)

Modern economy is moving towards higher fixed costs and lower marginal costs, which increases the incentives to price discriminate

Information goods (software, music): prototypical example

Also true of other high-tech products:

Pharmaceuticals

Microprocessors

Pentium prices: \$100–500

Marginal cost $\sim$ \$30

Communication satellites

Cars: design and tooling costs of \$2–3B for each new model

Conventional arguments for automated agents,
B2B exchanges, etc.:

better matching of supply and demand increases
efficiency

It appears that much greater gains can be
achieved through price discrimination

Wall Street Journal, June 8, 2001: Dell Latitude
L400 ultralight laptop listed at \$2,072.04, \$2,228,
and \$2,307 on Dell Web pages (designed for state
and local governments, small businesses, and
health-care companies, respectively).

Airlines and railroads: different services or price discrimination?

There was some doubt a century ago, when privacy (transferable tickets) limited what railroads could do, but today, it is clear that price discrimination is the main driving force behind airline yield management:

Fares offered at www.continental.com on Feb. 27, 2002:

Minneapolis to Newark, NJ on Wed., March 20, returning Fri., March 22: \$772.50

Minneapolis to Newark, NJ on Wed., March 20, returning Wed., March 27: \$226.50

Newark, NJ to Minneapolis on Fri., March 22, returning Fri., March 27: \$246.50

Price discrimination through versioning:

It is not because of the few thousand francs which would have to be spent to put a roof over the third-class carriages or to upholster the third-class seats that some company or other has open carriages with wooden benches. What the company is trying to do is to prevent the passengers who can pay the second class fare from traveling third class; it hits the poor, not because it wants to hurt them, but to frighten the rich. And it is again for the same reason that the companies, having proved almost cruel to the third-class passengers and mean to the second-class ones, become lavish in dealing with first-class passengers. Having refused the poor what is necessary, they give the rich what is superfluous.

Jules Dupuit, 1849

Major efficiency gain from price discrimination:
elimination of costs of versioning (financial and
others)

Politics makes for strange bedfellows, but so does
economics:

On average, for each dollar American
consumers pay for prescription drugs, the
Germans are paying 71 cents; the Swedes, 68
cents; the British, 65 cents; the French, 57
cents, and the Italians, 51 cents.

Unfortunately, U.S. policy allows the
pharmaceutical industry to maintain that
price disparity. ... It's a moral outrage that
Congress continues to allow millions of
elderly and chronically ill Americans to
suffer and die because they cannot afford the
inflated prices charged for pharmaceuticals.

Congressman Bernie Sanders
letter to *Barron's*
Feb. 28, 2000

Limitation on price discrimination: very strong negative consumer reactions

Zach: can supply report on digital cash for \$3,000

potential buyers and their willingness to pay:

- Alice: \$700
- Bob: \$800
- Charlie: \$900
- Donna: \$1,000

If Zach sells the report to Alice, Bob, Charlie, and Donna at \$650, \$750, \$850, and \$950, respectively, everybody will be better off.

However, if this is done transparently, he will likely have three enemies and an ingrate!

Behavioral economics: serious constraints on price discrimination, “dynamic pricing,” auctions, ..., because of negative public reactions:

Ultimatum game:

1. \$10 to be divided by Alice and Barbara
2. Alice proposes a split (for example, \$7 for Alice, \$3 for Barbara)
3. (a) Barbara accepts: each gets specified amount
(b) Barbara rejects: neither gets anything

Railroads in the 19th century: extremely important, widely hated

The drama was over. The fight of Ranch and Railroad had been wrought out to its dreadful close. ... Yes, the Railroad had prevailed. The ranchers had been seized in the tentacles of the octopus; the iniquitous burden of extortionate freight rates had been imposed like a yoke of iron.

“The Octopus”

Main complaint about railroad pricing:
discrimination

Interstate Commerce Act of 1887: first serious
federal regulation

Provisions:

- Rates to be “just and reasonable”
- Personal discrimination forbidden
- “Undue or unreasonable preference”
forbidden
- Charging more for short than long haul on
same line forbidden
- Pooling forbidden
- Rates to be published
- ...

Government regulation of railroads:

- traditional view: limiting Robber Barons' obscene profits
- revisionist view: regulatory capture by industry
- more accurate view: stability for industry and prices perceived to be fair for customers, but at the expense of efficiency

Explicit price discrimination desirable, but dangerous, but there are other ways to achieve the same goal:

Microsoft Office (for Windows 3.11)

components:

Access	\$225
Excel	\$225
Power Point	\$225
Word	\$175
	\$850

Office Pro bundle: \$389

Bundling is an alternative to price discrimination in reducing consumer surplus:

Willingness to pay:

	word processor	spreadsheet
Alice	\$100	\$300
Bob	\$300	\$100

Pricing and revenue:

\$100 for each program → \$400

\$300 for each program → \$600

\$400 for bundle → \$800

Incentives for sellers:

- price discriminate
- increase usage
- hide price discrimination

Most likely solutions: combination of techniques such as

- personalized bundles
- loyalty programs

Conclusions:

- Data mining to flourish, privacy to suffer
- Government role likely to be substantial, but ambiguous, since price discrimination is often socially desirable
- DRM, auctions, micropayments, anonymous digital cash, ..., to play minor role

References: several papers at

<http://www.dtc.umn.edu/~odlyzko>

especially “The bumpy road of electronic commerce,” *Proc. WebNet 96*, and others in preparation (“Privacy, price discrimination, and the future of ecommerce,” “Fairness and price discrimination on the Internet and on 19th century railroads,” and “Stronger copyright protection for cyberspace: Desirable, inevitable, and irrelevant”)

Also:

- Shapiro and Varian, *Information Rules*, 1998
- Bruce Schneier, *Secrets and Lies: Digital Security in a Networked World*, 2000
- Ross Anderson, *Security Engineering - A Guide to Building Dependable Distributed Systems*, 2001